



AI-Powered Flight Recovery Orchestrator

Transforming disrupted journeys into seamless self-service experiences.

Overview

SkyRecover AI is an intelligent flight recovery platform designed for regional airlines to automate passenger recovery during flight disruptions. Instead of overwhelming customer support centers, the platform proactively analyzes disruption scenarios, determines the best recovery strategy, and empowers passengers to resolve their journey through a modern self-service experience.

Unlike traditional airline chatbots, SkyRecover AI focuses on **decision orchestration, workflow automation**, and **AI-assisted recovery recommendations** rather than simple question answering.

Problem Statement

Weather disruptions, operational delays, and cancellations often generate a massive spike in customer support requests.

Passengers typically want answers to questions like:

- Is my flight cancelled?
- Can I rebook?
- Am I eligible for a refund?
- What compensation will I receive?
- Do I qualify for a hotel or meal voucher?
- Should I wait or choose another flight?

Traditional support centers become overloaded, leading to long wait times and poor customer satisfaction.

SkyRecover AI eliminates these bottlenecks by providing an intelligent self-service recovery platform.

Objectives

- Reduce customer support calls.
- Improve passenger experience.
- Automate recovery workflows.

- Recommend the best recovery option using AI.
 - Escalate only complex cases to human agents.
 - Provide transparent explanations for every decision.
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Key Features

Passenger Dashboard

- Secure passenger login
 - Upcoming journey information
 - Live flight status
 - Recovery timeline
 - Personalized recovery options
 - Notifications
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AI Recovery Assistant

Instead of waiting for passengers to ask questions, the AI proactively analyzes disruptions and recommends the optimal recovery plan.

Example:

Flight SJ238 has been cancelled due to severe weather. Based on your itinerary, loyalty status, and available seats, we recommend rebooking to Flight SJ241 departing at 18:45. Meal vouchers are included.

Intelligent Recovery Engine

The core of the platform.

Automatically determines:

- Cancellation recovery
 - Delay handling
 - Alternative flights
 - Refund eligibility
 - Voucher eligibility
 - Compensation
 - Priority handling
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Decision Engine

The decision engine combines:

- Airline business rules
- Passenger profile
- Flight availability
- Operational constraints
- AI recommendations

Instead of simply generating text, the system makes explainable business decisions.

Rule Engine

Handles deterministic airline policies.

Examples:

- Delay > 3 hours
- International flight
- VIP passenger
- Family booking
- Connecting flights
- Medical assistance
- No available seats

These rules execute before AI recommendations.

AI Recommendation Engine

Analyzes passenger context and generates the best recovery strategy.

Factors considered:

- Flight status
- Weather conditions
- Passenger loyalty
- Family bookings
- Alternate flights
- Estimated arrival
- Recovery cost
- Customer satisfaction

Outputs:

- Recommended action
 - Reasoning
 - Confidence score
 - Recovery summary
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Confidence Scoring

Every AI recommendation includes a confidence score.

Example

Confidence: 96%

Recommendation:
Auto-Rebook

Reason:
Alternative flight available with minimal delay.

Lower confidence recommendations automatically require human approval.

Example

Confidence: 48%

Action Required:
Escalate to Recovery Agent

Human Escalation

Certain scenarios remain agent-assisted.

Examples:

- International visa issues
- Special medical assistance
- Group bookings
- Manual compensation
- Low AI confidence

- Exceptional policies
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Notifications

Passengers receive updates via:

- Email
 - SMS
 - In-app notifications
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Supported Recovery Scenarios

Flight Cancelled

Passenger receives:

- Explanation
 - Alternative flights
 - Refund eligibility
 - Compensation
 - Hotel voucher (if applicable)
 - Meal voucher (if applicable)
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Flight Delayed

Passenger receives:

- Delay information
 - Updated boarding time
 - Voucher eligibility
 - Gate changes
 - Estimated arrival
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Missed Connection

Passenger receives:

- New itinerary
- Alternative connection
- Overnight accommodation

- Assistance request

Customer Journey

Passenger Login

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Journey Dashboard

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Flight Status Check

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Disruption Detected

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Recovery Engine

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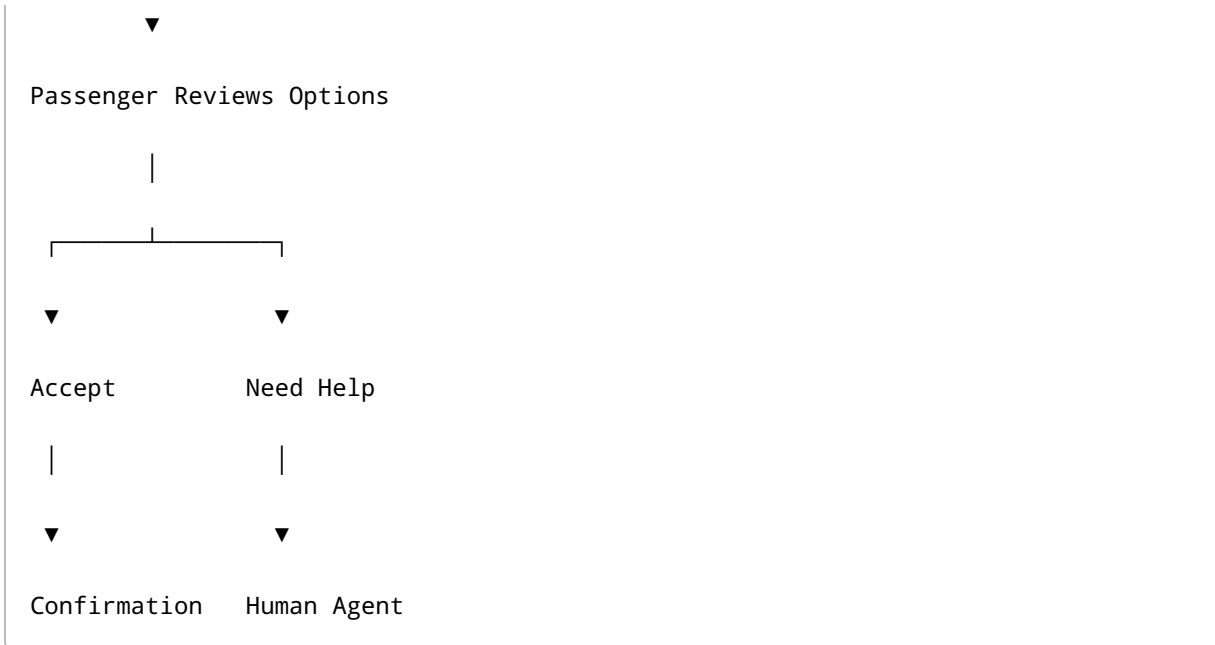
Decision Engine

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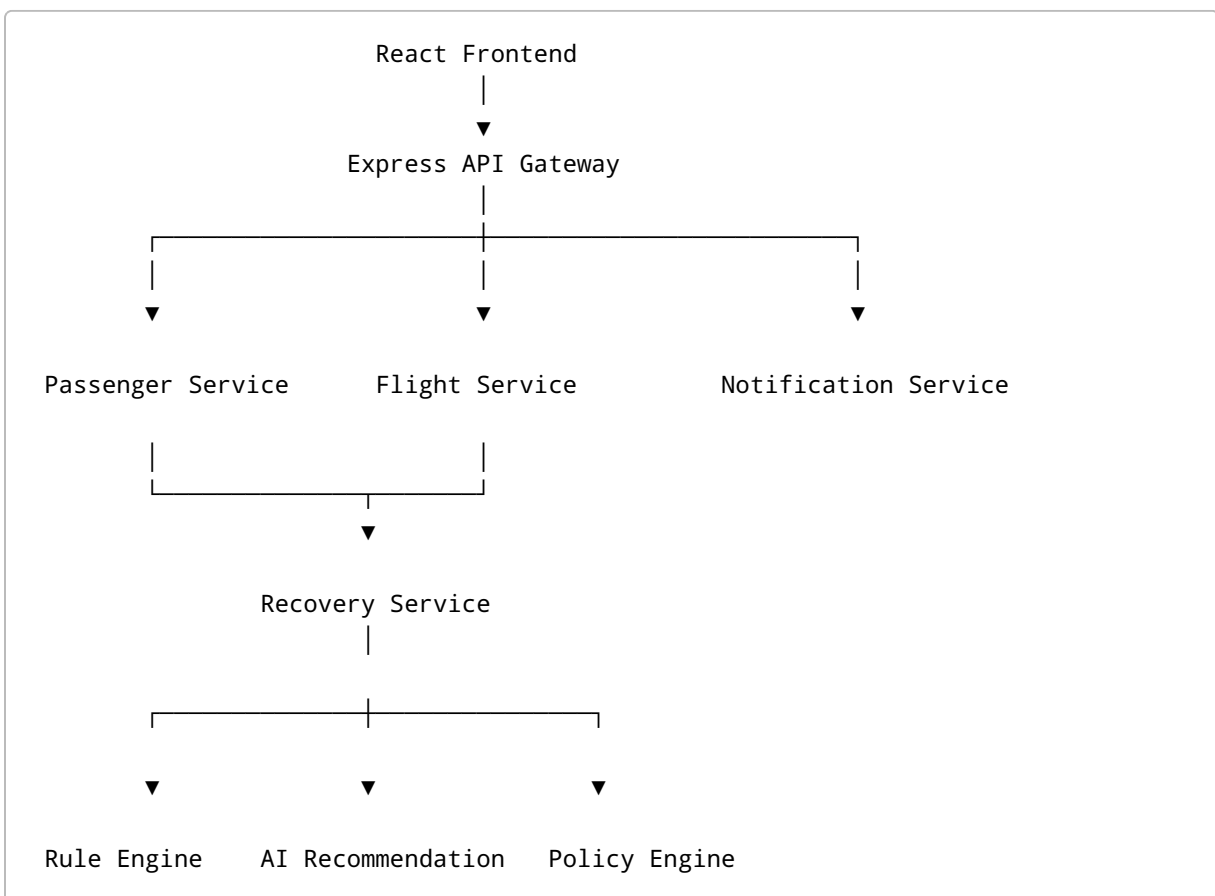


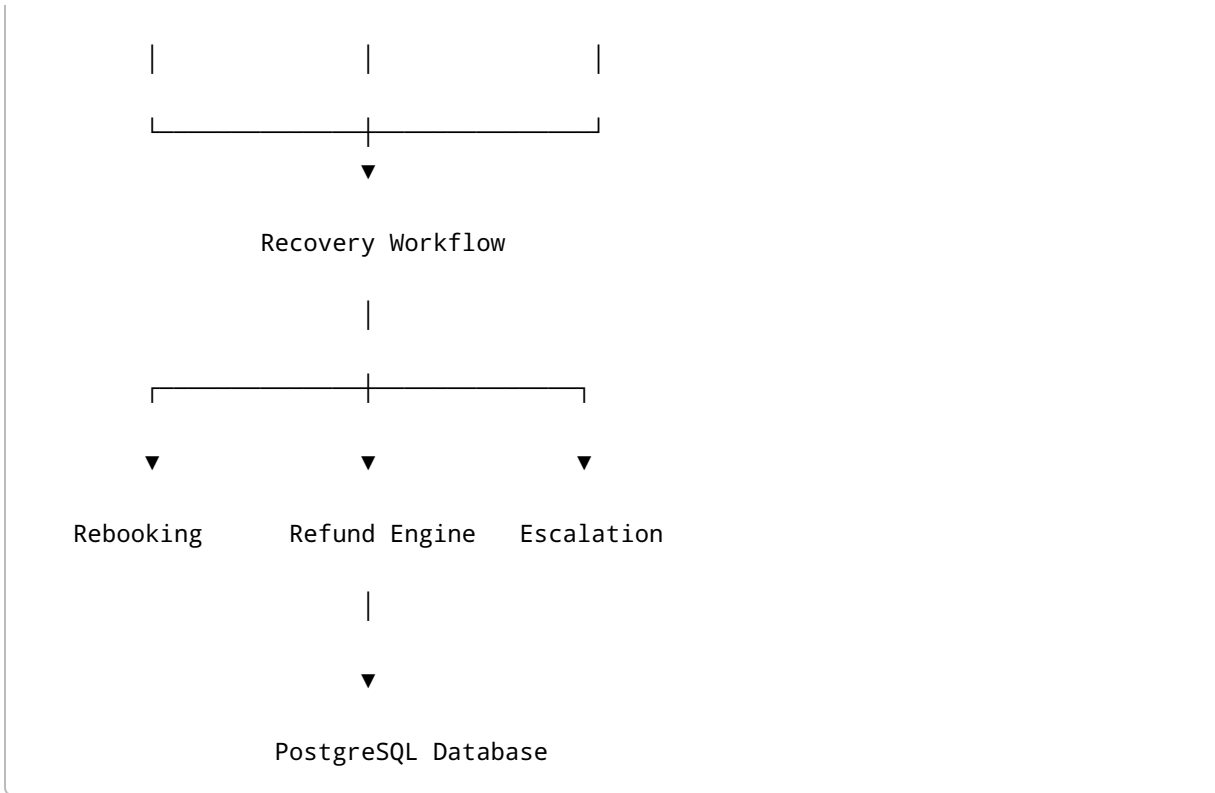
AI Recommendation

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High-Level Architecture





System Components

Frontend

- React
- Tailwind CSS
- Responsive Dashboard

Backend

- Node.js
- Express.js

Database

- PostgreSQL

Stores:

- Passengers
 - Flights
 - Recovery requests
 - Notifications
 - Recovery history
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AI Layer

Responsible for:

- Natural language explanations
 - Recovery recommendations
 - Passenger assistance
 - Confidence scoring
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Rule Engine

Implements airline policies.

Examples:

- Refund eligibility
 - Compensation rules
 - Voucher generation
 - Seat availability
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Recovery Workflow Engine

Coordinates:

- Rule Engine
 - AI Engine
 - Flight Service
 - Passenger Service
 - Notification Service
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Suggested REST API Design

Passenger

GET /api/passenger/:id

GET /api/passenger/:id/journey

Flights

GET /api/flights

GET /api/flights/:flightId

Recovery

POST /api/recovery/analyze

GET /api/recovery/options/:bookingId

POST /api/recovery/rebook

POST /api/recovery/refund

Notifications

POST /api/notifications/send

AI

POST /api/ai/recommend

POST /api/ai/chat

Database Tables

- passengers
 - bookings
 - flights
 - disruptions
 - recovery_requests
 - recovery_recommendations
 - notifications
 - vouchers
 - refunds
 - escalations
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Technology Stack

Frontend

- React
- Tailwind CSS

Backend

- Node.js
- Express.js

Database

- PostgreSQL

AI

- OpenAI API (or mocked LLM)
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Future Enhancements

- Live airline APIs
 - Real-time weather integration
 - Dynamic seat allocation
 - Loyalty program integration
 - Voice assistant
 - WhatsApp support
 - Predictive disruption alerts
 - Multi-language support
 - Analytics dashboard
 - Event-driven architecture using Kafka or RabbitMQ
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Key Assumptions

- Airline data is available through REST APIs.
 - Payments are out of scope.
 - Authentication is simplified.
 - Weather information is simulated.
 - Recovery policies are configurable.
 - AI recommendations assist decision-making but do not override mandatory airline rules.
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Why This Solution Stands Out

Unlike traditional airline chatbots, **SkyRecover AI** acts as an intelligent orchestration platform.

It combines:

- Business rules
- AI reasoning
- Workflow automation
- Explainable recommendations
- Confidence-based escalation
- Enterprise-grade architecture

This ensures passengers receive fast, transparent, and personalized recovery options while reducing operational workload for airlines.

Team

Built as a hackathon MVP to demonstrate how AI-powered orchestration can transform airline disruption management through intelligent self-service recovery.